Universität Bonn

Mathematisches Institut
Prof. Dr. Shuntaro Yamagishi
Number Theory I
Summer 2026

Lecture Notes as of 17th April 2026

LaTeX by Hannes Heimberger

Corrections and Remarks

If you find a mistake in this file or have a remark concerning this document, I would be glad if you let me know. It would be great if you wrote a mail. I can be reached through the address `hannes.heimberger@uni-bonn.de`.

Solutions to exercises

I will add solutions to the exercises after their correction, at least if my tutor marked them correct. These may, however, not be written down in the most optimal way. The proofs left as exercsises will be marked as such. If you have suggestions about how to improve these proofs, comments are welcome.

Colours

My annotations to the notes are blue. Additional details that are neither part of the lecture nor relevant to the exam will be coloured in orange.

Thanks

I also thank Daniel Emse for providing me with his template, which I adapted for these notes.

Contents

Number Theory I

1 Geometry of Numbers **2**
1.1 Introduction 2
1.2 Geometry of numbers 4

Appendix

List of Definitions **A**

List of Theorems **B**

List of Lemmata **C**

Index **D**

Number Theory I

Notations and conventions

Number sets

- The natural numbers, denoted by $\mathbb{N}$, contain 0 (which is in accord with *DIN 5473* (*German Industrial Norm*) – Welcome to Germany! xD).

Subsets and -structures

- If we write $A \subset B$, we mean $A \subseteq B$ and $A \neq B$ (see *DIN 5473*).
- We write $\sqsubset$ or $\sqsubseteq$ for substructures (subfields, linear subspaces of vector spaces, etc.) of algebraic structures (didn't yet make it into *DIN 5473* :().

For $f : \mathbb{C} \times \cdots \to \mathbb{C}$, we define

$$N(f) = \{s \in \mathbb{C} : f(s, -) = 0\}$$

and

$$N^*(f) = \{s \in [0, 1] \times \mathbb{R} \subset \mathbb{C} : f(s, -) = 0\}.$$

Quantors

- If we write $\widehat{\forall}_{\ldots}$, we mean something holds for almost all ...

Functions

- We write $(x, y) = \gcd(x, y)$.
- We have $\{x\} = x - \lfloor x \rfloor \ \ \forall_{x \in \mathbb{R}}$.

Chapter 1

Geometry of Numbers

1.1 Introduction

Definition 1.1.1 – $\mathbb{Z}^2_{\text{prim}}$

We denote

$$\mathbb{Z}^2_{\text{prim}} = \{(x, y) \in \mathbb{Z}^2 : (x, y) = 1\}.$$

Theorem 1.1.2 (Dirichlet, 1842)

Let $\alpha \in \mathbb{R}$. Then, for every integer $Q \geq 2$, there exists $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $0 < y \leq Q$ and $|x - \alpha y| \leq 1/Q$.

Proof. Consider the $Q+1$ numbers $0, \alpha - \lfloor \alpha \rfloor, 2\alpha - \lfloor 2\alpha \rfloor, \ldots, Q\alpha - \lfloor Q\alpha \rfloor$. Each of these is in the interval $[0, 1)$. Partition the interval $[0, 1)$ into Q subintervals of length $1/Q$. Now, we can apply the pigeonhole principle to deduce that

$$\exists_{0 \leq i \leq j \leq Q} : |(j\alpha - \lfloor j\alpha \rfloor) - (i\alpha - \lfloor i\alpha \rfloor)| \leq \frac{1}{Q}.$$

This implies

$$|(j - i)\alpha - (\lfloor j\alpha \rfloor - \lfloor i\alpha \rfloor)| \leq \frac{1}{Q}.$$

Let $\widetilde{y} = j - i$ and $\widetilde{x} = \lfloor j\alpha \rfloor - \lfloor i\alpha \rfloor$. Then $|\widetilde{y}\alpha - \widetilde{x}| \leq 1/Q$. Let $d = (\widetilde{y}, \widetilde{x})$, $y = \widetilde{y}/d$ and $x = \widetilde{x}/d$ and divide through by d to get that

$$|y\alpha - x| \leq \frac{1}{dQ} \leq \frac{1}{Q}.$$

□

A very simple consequence of this:

Corollary 1.1.3

Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. Then there exist infinitely many $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $y > 0$ and

$$\left| \alpha - \frac{x}{y} \right| \leq \frac{1}{y^2}.$$

Proof. For each $Q \geq 2$, let $(x_Q, y_Q) \in \mathbb{Z}^2_{\text{prim}}$ be such that $0 < y_Q \leq Q$ and $|x_Q - y_Q\alpha| \leq {}^1\!/_Q$, which we know to exist from Theorem 1.1.2. This implies

$$\left|\alpha - \frac{x_Q}{y_Q}\right| \leq \frac{1}{Qy_Q} \leq \frac{1}{y_Q^2}.$$

Suppose the set

$$S = \{(x_Q, y_Q) : Q \geq 2\}$$

is finite. Then there exists $(x_0, y_0) \in \mathbb{Z}^2_{\text{prim}}$ such that $(x_Q, y_Q) = (x_0, y_0)\ \widehat{\forall}_{Q \geq Q_0}$ for some Q_0. This means

$$\left|\alpha - \frac{x_Q}{y_Q}\right| = \left|\alpha - \frac{x_0}{y_0}\right| \leq \frac{1}{Q}\ \widehat{\forall}_{Q \geq Q_0}.$$

Taking the limit $Q \to \infty$ gives

$$\alpha = \frac{x_0}{y_0},$$

which is a contradiction due to the fact that α was supposed to be rational. ↯ □

Question 1.1.4

Can we do better than ${}^1\!/_{y^2}$ for some $\alpha \in \mathbb{R}^2$? I.e. do there exist infinitely many $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $y > 0$ and

$$\exists_{\delta > 0} : \left|\alpha - \frac{x}{y}\right| < \frac{1}{y^{2+\delta}}.$$

Answer. Yes, but it's rare. However, this is not the case if α is algebraic over $\mathbb{Q}$. □

Theorem 1.1.5 (Roth, 1955)

Let $\alpha \in \mathbb{A}$ and $\delta > 0$. Then

$$\#\{(x, y) \in \mathbb{Z}^2_{\text{prim}} : y > 0, |\alpha - {}^x\!/_y| \leq {}^1\!/_{y^{2+\delta}}\} < \aleph_0.$$

Proving this is one of the main goals of this course. Another important result we will prove is Thue's theorem.

Theorem 1.1.6 (Thue, 1909)

Let $F \in \mathbb{Z}[X, Y]$ be a binary form (hence homogenious) of degree d. Suppose $\deg F(X, 1) = d$ and that $F(X, 1)$ has at least three distinct zeroes in $\mathbb{C}$. Let $m \in \mathbb{Z} \setminus \{0\}$. Then

$$\#\{(x, y) \in \mathbb{Z}^2 : F(x, y) = m\} < \aleph_0.$$

Also, time permitted, Professor Yamagishi'd like to go through the details of Stanley Yao Xiao's proof of Büchi's problem.

Problem 1.1.7 (Büchi)

Let $0 < x_1 < \cdots < x_5$ be integers such that

$$x_{i+1}^2 - 2x_i^2 + x_{i-1}^2 = 2 \ \forall_{i \in \{2,\ldots,4\}}.$$

Then there exists $x_0 \in \mathbb{Z}$ such that $x_i = x_0 + i \ \forall_{1 \leq i \leq 5}$.

This leads to the following refinement of Hilbert's tenth problem: There is no algorithm which decides the existence of integer solutions to an arbitrary system of diagonal quadratic equations:

$$A \begin{pmatrix} x_1^2 \\ \vdots \\ x_n^2 \end{pmatrix} = b.$$

Professor Yamagishi: This is a paper by a friend of mine. Actually, it has nothing to do with geometry of numbers. But I want to do it anyway.
Do lecture subtitle!
Professor Yamagishi recommends the book **An introduction to the geometry of numbers** by Cassels. He also recommend a course on **Prime Numbers** taught next semester (Selected Topics in Number Theory). Tutorial times are **Monday at 14:00** and **Tuesday at 12:00**. You can register for these **Tomorrow at 16:00**.

Lecture 1 (15th April 2026)

. . . . . .

Lecture 2 (17th April 2026)

1.2 Geometry of numbers

Let $n \geq 2$. Given $x_1, \ldots, x_n \in \mathbb{R}^n$, we say that they are linearly independent over $\mathbb{R}$, if $c_1x_1 + \cdots + c_rx_r \neq 0 \ \forall_{(c_1,\ldots,c_r)^t \in \mathbb{R}^r \setminus \{0\}}$. We consider the usual topology on $\mathbb{R}^n$ generated by the Euclidean metric. For subsets of $\mathbb{R}^n$, we consider the subspace topology induced by this: We say a subset $\Lambda \subseteq \mathbb{R}^n$ is **discrete** if $\#\Lambda \cap B < \aleph_0$ for every bounded subset $B \subseteq \mathbb{R}^n$. Furthermore, Λ is a **discrete subgroup** if it is discrete and $c_1x_1 + c_2x_2 \in \Lambda \ \forall_{c_1,c_2 \in \mathbb{Z}, x_1,x_2 \in \Lambda}$. In this case, we define $\mathrm{rank}(\Lambda) = r$ as the maximal number r such that Λ contains r linearly independent vectors over $\mathbb{R}$. A **basis** of Λ is a set $\{v_1, \ldots, v_r\}$ of linearly independent vectors over $\mathbb{R}$ such that $\Lambda = \{c_1v_1 + \cdots + c_rv_r : c_1, \ldots, c_r \in \mathbb{Z}\}$.

Lemma 1.2.1

Let Λ be a discrete subgroup of $\mathbb{R}^n$ and $\mathrm{rank}(r) = r \geq 1$. Also assume that $\{w_1, \ldots, w_r\} \subseteq \Lambda$ is linearly independent over $\mathbb{R}$. Then there exists a basis $\{v_1, \ldots, v_r\}$ of Λ such that for all $1 \leq k \leq r$, we have

$$v_k = \alpha_k w_1 + \cdots + \alpha_k w_k \ \forall_{\alpha_{k,1},\ldots,\alpha_{k,k} \in [0,1] \text{ and } \alpha_{k,k} \neq 0}.$$

Proof. Let $1 \leq k \leq r$. We define

$$S_k = \{\alpha_1 w_1 + \cdots + \alpha_k w_k : a_1, \ldots, a_k \in [0,1], \alpha_k \neq 0\}.$$

Since Λ is discrete and S_k is bounded,

$$\underbrace{0 <}_{w_k \in \Lambda \cap S_k} \#(\Lambda \cap S_k) \ll 1.$$

We choose v_k to be a vector from $\Lambda \cap S_k$ with the minimal α_k. Let us write

$$v_k = \alpha_{k,1} w_1 + \cdots + a_{k,k} w_k.$$

Then for any

$$\alpha_1 w_1 + \cdots + a_k w_k \in \Lambda \cap S_k,$$

we have $0 < \alpha_{k,k} \leq \alpha_k \leq 1$. It is clear that $\{v_1, \ldots, v_r\}$ is linearly independent over $\mathbb{R}$. For each $0 \leq k \leq r$, let

$$M_k = \Lambda \cap \left\{ \sum_{j=1}^{k} \alpha_j w_j : \alpha_1, \ldots, \alpha_k \in \mathbb{R} \right\} \text{ with } M_0 = \{0\}.$$

Note that $M_r = \Lambda$. We now use induction to show that $M_k = \{c_1 v_1 + \cdots + c_k v_k : c_1, \ldots, c_k \in \mathbb{Z}\}$ $\forall_{0 \leq k \leq r}$. When $k = 0$, the statement is trivially true. Suppose the statement holds for $k - 1$.
Let

$$x = \sum_{j=1}^{k} \beta_j w_j \in M_k \subseteq \Lambda \text{ and set } z_k = \left\lfloor \frac{\beta_k}{\alpha_{k,k}} \right\rfloor \in \mathbb{Z}.$$

Then, $0 \leq \beta_k - z_k \alpha_{k,k} < \alpha_{k,k}$. Consider $x - z_k \underbrace{v_k}_{\in \Lambda} \in \Lambda$. Also,

$$\begin{aligned} x - z_k v_k &= \sum_{j=1}^{k} (\beta_j - z_k \alpha_{k,j}) w_j \\ &= \underbrace{\sum_{j=1}^{k} \lfloor \beta_j - z_k \alpha_{k,j} \rfloor w_j}_{\in \Lambda} + u, \end{aligned}$$

where

$$u = \sum_{j=1}^{k-1} \{\beta_j - z_k \alpha_{k,j}\} w_j + (\beta_k - z_k \alpha_{k,k} w_k).$$

It is clear that $u \in \Lambda$. Also, if $\beta_k - z_k \alpha_{k,k} > 0$, then $u \in S_k$ (because $\{\beta_j - z_k \alpha_{k,j}\} \in [0, 1]$ and $0 < \beta_k - z_k \alpha_{k,k} < \alpha_{k,k} \leq 1$), which implies $u \in \Lambda \cap S_k$. But this contradicts the minimality of $\alpha_{k,k}$ $\lightning$. Hence, $\beta_k = z_k \alpha_{k,k}$. Then,

$$x - z_k v_k = \sum_{j=1}^{k-1} (\beta_j - z_k \alpha_{k,j}) w_j \in M_{k-1}.$$

This concludes the induction. $\square$

Definition 1.2.2 – (Full) Lattice

A **(full) lattice** in $\mathbb{R}^n$ is a discrete sugroup of $\mathbb{R}^n$ which has rank n.

Remark (Bases of Lattices)**:** If Λ is a lattice in $\mathbb{R}^n$, then by the previous lemma, there exists a basis $\{v_1, \dots, v_n\}$ such that $\Lambda = \{c_1 v_1 + \cdots + c_n v_n : c_1, \dots, c_n \in \mathbb{Z}\}$.

Definition 1.2.3 – Determinant of a Lattice

We define the **determinant** of Λ to be

$$\det(\Lambda) = |\det(v_1 \cdots v_n)|.$$

Recall that $\mathrm{GL}_n(\mathbb{Z})$ is the multiplicative group of $n \times n$-matrices over $\mathbb{Z}$ with determinant ± 1.

Lemma 1.2.4

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Suppose that $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_n\}$ are two bases of Λ. Then there exists $u = (u_{i,j})_{i,j \in \{1,\dots,n\}} \in \mathrm{GL}_n(\mathbb{Z})$ such that

$$w_i = \sum_{j=1}^{n} n_{i,j} v_j \ \forall_{1 \le i \le n}.$$

In particular,

$$|\det(w_1 \cdots w_n)| = |\det(v_1 \cdots v_n)|.$$

Proof. Let us define $U = (u_{i,j})_{i,j \in \{1,\dots,n\}}$ by

$$w_i = \sum_{j=1}^{n} u_{i,j} v_j \ \forall_{1 \le i \le n}.$$

Since $w_i \in \Lambda$ and $\{v_1, \dots, v_n\}$ is a basis of Λ, we get $u_{i,j} \in \mathbb{Z}$ $\forall_{1 \le i,j \le n}$, which ensures $\det(U) \in \mathbb{Z}$. By the same argument, we define $\widetilde{U} = (\widetilde{u}_{i,j})_{i,j \in \{1,\dots,n\}}$ analogously. Note that $U^{-1} = \widetilde{U}$. Hence, $\det(\widetilde{U}) = \det(U) = \pm 1$. $\square$

Let M, L be two lattices in $\mathbb{R}^n$ with $M \subseteq L$ and. Further let $\{v_1, \dots, v_n\}$ be a basis of L and $\{w_1, \dots, w_m\}$ be a basis of M. Define U_0 as the $n \times n$ matrix that fulfils

$$(w_1 \cdots w_n) = (v_1 \cdots v_n) U_0.$$

Then, we have $U_0 \in M_n(\mathbb{Z})$. We define the **index** of M in L by

$$|L : M| = |\det U_0| = \frac{\det M}{\det L}.$$

Note: This does not depend on the choice of bases of M and L.

Example. Let

$$\Lambda_0 = \mathrm{Span}_{\mathbb{Z}} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \text{ and } \Lambda = \mathrm{Span}_{\mathbb{Z}} \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}. \qquad \square$$

Appendix

List of Definitions

1.1.1 $\mathbb{Z}^2_{\text{prim}}$ 2
1.2.2 (Full) Lattice 6
1.2.3 Determinant of a Lattice 6

List of Theorems

1.1.2 Dirichlet, 1842 2
1.1.5 Roth, 1955 3
1.1.6 Thue, 1909 3

List of Lemmata

1.2.1 4
1.2.4 6

Index

$N(f)$, 1
$N^*(f)$, 1
$\mathbb{Z}^2_{\text{prim}}$, 2
(Full) Lattice, 6

Basis of a Discrete Subgroup of $\mathbb{R}^n$, 4

Determinant of a Lattice, 6
Discrete Subgroup of $\mathbb{R}^n$, 4
Discrete Subset of $\mathbb{R}^n$, 4

Index of a Sublattice in a Lattice, 6

Rank of a Discrete Subgroup of $\mathbb{R}^n$, 4